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4 Cystic fibrosis is an inherited recessive disease caused by mutations in a gene on chromosome 7.

(a) Give the meaning of the term **gene**.

(1)

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(b) Explain how mutations result in cystic fibrosis.

(3)

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(c) Population carrier screening (PCS) is one type of genetic screening.

This involves screening people who want a child to see if they are carriers of genetic disorders.

In one country, the number of babies born with cystic fibrosis went down following the introduction of PCS.

Suggest why the number of babies born with cystic fibrosis went down.

(3)

(Total for Question 4 = 7 marks)

